

Karan Uppal

 Website |  karan.uppal3@gmail.com |  Google Scholar |  LinkedIn

EDUCATION

University of California Los Angeles

2026 – Present

PhD in Computer Science

Indian Institute of Technology (IIT) Kharagpur, India

2019 – 2024

Combined BSc. & MSc. in Mathematics and Computing

GPA: 9.16/10

SELECTED PUBLICATIONS

* denotes equal contribution

MoVE: Mixture-of-Vocabulary-Experts for Improved Representation Learning

Under review

Karan Uppal, Nagarajan Natarajan, Manik Varma



Understanding Task Transfer in Vision-Language Models

CVPR '26 (Oral)

Bhuvan Sachdeva*, Karan Uppal*, Abhinav Java*, Vineeth N. Balasubramanian



Swift Sampling: Selecting Temporal Surprises via Taylor Series

NeurIPS '26

Dahye Kim, Bhuvan Sachdeva*, Karan Uppal*, Naman Gupta*, Vineeth N. Balasubramanian, Deepti Ghadiyaram



RESEARCH EXPERIENCE

Research Fellow, Microsoft Research India

Jul 2024 – Jun 2026

Guide: Dr. Nagarajan Natarajan, Prof. Vineeth N Balasubramanian, Prof. Manik Varma

Bengaluru, India

- Scaled vocabulary sizes beyond 1M using Mixture-of-Experts to pretrain dense retrieval models for Bing Ads
- Investigated task transferability in Vision-Language Models to uncover cross-task interactions

DAAD WISE Scholar | Max Planck Institute for Intelligent Systems

May 2022 – Jul 2022

Guide: Dr. Shashank Singh, Prof. Bernhard Schölkopf

Tübingen, Germany

- Released a new dataset and baseline models to decode human attention from gaze data in road scenes
- Designed a novel CNN architecture to incorporate temporal gaze data for object localization

Research Intern | Harvard University

May 2021 – Aug 2021

Guide: Prof. Hanspeter Pfister

Cambridge, USA

- Experimented with different variants of the 3D UNet model for instance segmentation on brain volumes
- Achieved 1st position on the CREMI leaderboard for synaptic cleft detection task using self-learning

WORK EXPERIENCE

AI Resident | Ema Unlimited

Jan. 2024 – Apr 2024

Early stage startup for Agentic AI-powered employees

Bengaluru, India

- Created an agentic pipeline to convert natural language to SQL queries over enterprise databases
- Developed multi-LLM RAG-based ensemble framework for improved inference and accuracy

Summer Fellow | University of Warwick and UN-REDD

Jun 2023 – Aug 2023

Data Science for Social Good (DSSG)

Coventry, UK

- Implemented a spatiotemporal UNet for forecasting deforestation in the Brazilian Amazon
- Deployed a low-latency visualization tool with feature ablation to analyze drivers of deforestation

Applied Research Intern | NVIDIA

Aug 2021 – Jan 2022

WWFO - Solution Architect and Engineering

Bengaluru, India

- Developed and deployed 5 scalable pipelines for the tasks of image classification and object detection
- Achieved 10x speed-up in model inference for retrieval, re-ranking and QnA end-to-end applications

COMPETITIONS AND CHALLENGES

American Express Data Science Challenge

Sep 2022 – Nov 2022

One of India's largest Data Science Competitions

Gurugram, India

- Achieved 2nd place out of 3600+ teams for the task of predicting credit card customer default
- Attained 82.4% accuracy using a gradient boosting ensemble, battling dataset imbalance

Machine Learning Reproducibility Challenge

Nov 2021 – Feb 2022

Papers with Code Challenge

Remote

- Replicated and extended a CVPR 2021 publication for reproducibility of its claims; presented at NeurIPS '22
- Implemented Noise-Aware Loss from scratch achieving state-of-the-art results on PASCAL VOC 2012

IROS-RSJ Navigation and Manipulation Challenge

Jul 2021 – Sep 2021

Multiple waypoint navigation with obstacles

Prague, Czech Republic

- Attained 1st place for the task of navigation and manipulation of a robot in unknown indoor environments
- Developed a probabilistic distance minimization algorithm for path planning; published at ICCR 2022

ACHIEVEMENTS

TSG, IIT KGP	Awarded Honorable Mention for distinction in technology endeavours in campus	2024
DSSG Fellow	Selected internationally to pursue a project with the UN-REDD in Coventry, UK	2023
DAAD WISE	Awarded a €3000 grant to pursue research in Germany as a visiting researcher	2022
INSPIRE Scholar	Awarded 60,000 INR by the Government of India for excellence in sciences	2021
JEE Main	Attained a percentile of 99.87% among more than 1.5 million applicants	2019
JEE Advanced	Ranked in the top 0.82% among the 200k shortlisted candidates from above	2019

SKILLS

ML/AI: PyTorch (Lightning), Accelerate, Transformers, WandB, Hydra
Programming languages: C/C++, Python, Bash, SQL, Javascript

Coursework: Transcript [↗](#)
Misc.: Docker, Git, LaTeX, Markdown

POSITIONS OF RESPONSIBILITY

Volunteership

Aug 2022 – Nov 2022

Independent Endeavors

IIT Kharagpur

- Organized a donation drive for an orphanage in IIT Kharagpur, raising over 55,000 INR
- Conducted a blood donation drive with 120+ participants across the campus

Deep Learning Team Lead

Mar 2021 – May 2022

Autonomous Ground Vehicle Research Group

IIT Khargpur

- Headed a team of 6 sophomores for participation in Machine Learning Reproducibility Challenge 2021
- Organized over 10 sessions for a team of 15 freshers, for training in Image Processing and Deep Learning

Head Analytics

Jul 2021 – May 2022

Business Club

IIT Kharagpur

- Organized the 4th international edition of our flagship event, witnessing 3000+ participation
- Conducted 22 bi-weekly knowledge sessions for a team of 30+ members on the topics of Machine Learning

EXTRACURRICULARS

Open Source : Contributed to the huggingface/transformers repository for bug fixes in the MoE architecture

Reproducible Science : Authored manuscript for the ML Reproducibility Challenge, presented at NeurIPS 2022

Technical Writing : Writer of a blog series summarizing fundamental research papers in the field of deep learning

Teaching : Mentored children from the economically weaker section of 6th to 8th standards in academic activities